

Multiple Choice (10 marks)

1. To verify a digital signature we need the
 - (a) Sender's Private key
 - (b) Sender's Public key
 - (c) Receiver's Private key
 - (d) Receiver's Public key
2. Which of the following deals with network intrusion detection and real-time traffic analysis?
 - (a) John the Ripper
 - (b) L0phtCrack
 - (c) Snort
 - (d) Nessus
3. The digest created by a hash function is normally called a
 - (a) Modification detection code (MDC)
 - (b) Modify authentication connection
 - (c) Message authentication control
 - (d) Message authentication cipher
4. What does EXE do when the STP solver times out on a constraint query for a particular path?
 - (a) Assume that the query is satisfiable and continue executing the path.
 - (b) Assume that the query is not satisfiable and stop executing the path
 - (c) Restart STP and retry the query, up to a limited number of retries.
 - (d) Remove a subset of the constraints and retry the query.
5. Can a stream cipher have perfect secrecy?
 - (a) Yes, if the PRG is really "secure"
 - (b) No, there are no ciphers with perfect secrecy
 - (c) Yes, every cipher has perfect secrecy
 - (d) No, since the key is shorter than the message

True / False (10 marks)

Answer True or False

6. True or False: A man-in-the-middle attack cannot affect the Diffie-Hellman method if the two parties are joined in a secure network.
7. True or False: The sub key length at each round of DES is 56 bits.
8. True or False: SQL queries based on user input are particularly vulnerable to injection attacks due to improper sanitization of user data.
9. True or False: EtterPeak is a widely used tool for network analysis in multiprotocol diverse networks.
10. True or False: The four primary security principles related to messages are Confidentiality, Access Control, Integrity, and Non-repudiation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function that matches a string that has an a followed by zero or one 'b'.
12. Write a function to check whether the given string starts and ends with the same character or not using regex.

End of Paper